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Directional lighting device

Abstract

The disclosure proposes an improved directional lighting device (**10**) comprising a supporting structure (**11**); a ball-joint structure (**12**) mounted to the supporting structure (**11**); a housing (**13**) configured to accommodate a light emitting source (**14**); electric wiring (**15**) fed from the supporting structure (**11**) through the ball-joint structure (**12**) and electrically connecting the light emitting source (**14**); the ball-joint structure (**12**) configured to direct the housing (**13**) relative to the supporting structure (**11**) and the light emitting source (**14**).

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Background/Summary

CROSS-REFERENCE TO PRIOR APPLICATIONS

(1) This application is the U.S. National Phase application under 35 U.S.C. § 371 of International Application No. PCT/EP2022/079568, filed on Oct. 24, 2022, which claims the benefit of European Patent Application No. 21205255.9, filed on Oct. 28, 2021. These applications are hereby incorporated by reference herein.

TECHNICAL FIELD

(2) The present disclosure relates to a lighting device with a directional lighting housing.

BACKGROUND OF THE INVENTION

(3) Directional lighting devices are commonly known in several configurations for illuminating

spaces in home and office environments. In general, such directional lighting devices have a supporting structure for mounting to a wall surface or ceiling surface and a lamp housing in which a light emitting source is accommodated. The lamp housing is connected to the supporting structure by means of a ball joint rotation system. Electric wiring is fed through the supporting structure, through the ball joint rotation system electrically connected with the light emitting source. The electric wiring at the supporting structure side is subsequently electrically connected with the power mains of the home and office environment.

(4) Several configurations employs power circuitry (power electronics) either in supporting structure or on the lamp housing. For directing the light beam of the lighting device, the ball joint rotation system allows the lamp housing with the light emitting source to be rotated in all directions around the ball surface of the ball joint. Continuous rotation around 360° causes twisting of the electric wiring, ultimately leading to rupture, malfunction and potential hazard due to electrical short circuit.

(5) Accordingly, it is a goal of the present disclosure to improve the presently known directional lighting devices to the extent that the improved configuration does not suffer from undesired twisting of the electric wiring when the housing is rotated around 360° multiple times and thus is not prone to rupture and potential electrical hazards.

SUMMARY OF THE INVENTION

(6) According to a first example of the disclosure, a directional lighting device is proposed, which comprises a supporting structure; a ball-joint structure mounted to the supporting structure; a housing configured to accommodate a light emitting source; electric wiring fed from the supporting structure through the ball-joint structure and electrically connecting the light emitting source; and wherein the ball-joint structure configured to direct the housing relative to the supporting structure and the light emitting source. As the lamp housing can be rotated in all directions around the ball surface of the ball joint structure relative to the supporting structure and the light emitting source no undesired twisting of the electric wiring occurs. Accordingly, the electric wiring within the lamp housing will not rupture and potential hazards, such as electric short-circuit is avoided.

(7) According to the disclosure, the light emitting source is mounted to a mounting plate, wherein the mounting plate has an outer dimension which is smaller than the inner dimension of the housing. In a particular example, the mounting plate is retained within the housing by means of a retaining spring element, which fits in a circumferential slot provided at the inner circumferential surface of the housing. As the light emitting source is kept stationary relative to the lamp housing, the latter rotation and directing in all directions around the ball surface of the ball joint structure does not twist the electric wire and accordingly any wire rupture is prevented.

(8) To further improve the positioning of the lamp housing relative to the ball-joint structure, a cup-shaped tensioning element having an outer dimension which is smaller than the inner dimension of the housing is clamped between the ball-joint structure and the mounting plate. The clamping of the cup-shaped tensioning element on the ball surface of the ball-joint structure is effected by the retaining spring element fitted in the circumferential slot provided in the inner circumferential surface of the housing. The clamping maintains a constant tension on the ball joint, and accordingly allows the lamp housing to be kept in the chosen position relative to the supporting structure.

(9) In a preferred example, the housing is configured as an elongated hollow tube with one tube end being mounted to the ball-joint structure. In particular, a bearing ring is mounted between the ball joint structure and the inner surface of the one tube end. Herewith the lamp housing can be directed and rotated properly in all directions relative the supporting structure and the light emitting source, without any risk of unwanted wire twisting and wire rupture.

(10) By mounting a tensioning spring between the cup-shaped tensioning element and the mounting plate the clamping can be maintained whilst allowing rotation and orientation of the lamp housing along the ball surface in any desired position.

(11) In particular, the cup-shaped tensioning element is provided with a circumferential slit for

accommodating the tensioning spring. Herewith a compact inner construction is obtained with limited risk for malfunction or blocking.

(12) In order to properly accommodate the electric wiring within the lamp housing without any further risk of twisting or entrapment between the several component inside the lamp housing, a flexible wiring sleeve is used in which the electric wiring can be accommodated. Preferably, the flexible wiring sleeve has a first sleeve end which is loosely coupled with the ball-joint structure and a second sleeve end which is loosely coupled with the mounting plate. The loose coupling of the flexible wiring sleeve within the directional lighting device prevents any twisting or entrapment during the rotation and direction of the lamp housing relative to the supporting structure and the light emitting device accommodated inside the housing.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The invention will now be discussed with reference to the drawings, which show in:

(2) FIG. 1 side views of a non-exemplary example of a directional lighting device according to the disclosure;

(3) FIGS. 2a and 2b sectional views of a non-exemplary example of a directional lighting device according to the disclosure;

(4) FIG. 3 a further detailed view of a non-exemplary example of a directional lighting device according to the disclosure.

DETAILED DESCRIPTION OF THE INVENTION

(5) For a proper understanding of the invention, in the detailed description below corresponding elements or parts of the invention will be denoted with identical reference numerals in the drawings.

(6) FIG. 1 depicts side views of a non-exemplary example of a directional lighting device according to the disclosure. Proper sectional views of the directional lighting device are depicted in FIGS. 2a-2b and 3. The directional lighting device is denoted with reference numeral **10** and is composed of a supporting structure **11**, a ball-joint structure **12** mounted to the supporting structure **11**, a housing **13** configured to accommodate a light emitting source **14** and electric wiring **15**. The electric wiring **15** is fed from the supporting structure **11** through the ball-joint structure **12** and is electrically connected with the light emitting source **14**.

(7) Such directional lighting device finds many applications for illuminating a spaces in home and office environments. In general, the supporting structure **11** is mounted to a wall surface **1** or ceiling surface **1** of the space to be illuminated using known construction materials. The electric wiring **15** at the supporting structure side is subsequently electrically connected with the power mains of the home or office environment.

(8) As shown in the sectional views of FIGS. 2a and 2b, the electric wiring **15** is fed through suitable feedthrough openings **11a** and **12a** of either supporting structure **11** and ball-joint structure **12** before being electrically connected to the light emitting source **14**. The supporting structure **11** is provided with a mounting flange **11b** which interacts with a mounting flange **12b** of the ball-joint structure **12**. Reference numeral **12z** denotes the main axis of ball-joint structure **12**. The main axis **12z** can be regarded as the source axis of the ball-joint structure **12** around which source axis the lamp housing **13** can be orientated and rotated around 360° relative to the supporting structure **11** (wall or ceiling surface **1**).

(9) In the state of the art directional lighting devices, continuous rotation around 360° causes twisting of the electric wiring, ultimately leading to rupture, malfunction and potential hazard due to electrical short circuit. The present disclosure provides an improved directional lighting device wherein the electric wiring **15** does not twist when the housing is rotated around 360° multiple

times and according to the electric wiring **15** is not prone to rupture and potential electrical hazards. (10) Accordingly, the ball-joint structure **12** is configured to direct the housing **13** relative to the supporting structure **11** and the light emitting source **14**. As the lamp housing **13** can be rotated in all directions around the ball surface of the ball joint structure **12** relative to the supporting structure **11** and the light emitting source **14** no undesired twisting of the electric wiring **15** occurs. The light emitting source **14** is kept stationary relative to the lamp housing **13**. Accordingly, the electric wiring **15** within the lamp housing **13** will not twist nor rupture and potential hazards, such as electric short-circuit is avoided.

(11) As shown in the Figures, the housing **13** is shaped as an elongated hollow tube with one tube end **13a**, serving as mounting end for mounting to the ball-joint structure **12**. The other tube end **13b** functions as light exit window and directs a light beam emitting by the light emitting source **15** along the longitudinal axis **13z** of the tube-shaped lamp housing **13**. The longitudinal axis **13z** can be regarded as the destination axis of the ball-joint structure **12** as it is pointing towards the destination of the projected light beam exiting the light exit window **13b**. For guaranteeing a proper mounting of the lamp tube housing **13** to the ball-joint structure **12** a bearing ring **22** is mounted between the ball joint structure **12** and the inner surface **13c** of the tube end **13a**. Accordingly, the lamp housing **13** can be directed and rotated properly in all directions relative the supporting structure **11** and the light emitting source **14**, without any risk of unwanted wire twisting and wire rupture.

(12) As stipulated, the configuration of the directional lighting device **10** according to the disclosure allows the light emitting source **14** to be kept stationary relative to the lamp housing **13**. To this effect, the light emitting source **14** is mounted to a mounting plate denoted with reference numeral **18**. The mounting plate **18** together with the light emitting source **14** are accommodated in the lamp housing **13**, with the mounting plate **18** facing the ball-joint structure **12** and the light emitting source **14** facing or directing towards the light exit window **13b** of the lamp housing **13**.

(13) The mounting plate **18** has an outer dimension which is conformal yet somewhat smaller than the inner dimension of the housing **13**, thus allowing the positioning of the mounting plate **18** in the inner space **130** of the tube-shaped housing **13**. In particular, the mounting plate **18** is loosely mounted inside the housing **13**, wherein it is retained within the housing **13** by means of a retaining spring element **19**. The retaining spring element **19** fits in a circumferential slot **13d** provided in the inner circumferential surface **13c** of the housing **13** and prevents the mounting plate **18** together with the light emitting source **14** from dislodging from the inner space **130** of the housing **13**.

(14) Accordingly, as the mounting plate **18** together with the light emitting source **14** are loosely mounted inside the housing **13**. In particular they are kept stationary relative to the lamp housing **13**, the latter rotation and directing in all directions around the ball surface of the ball joint structure **12** does not twist the electric wire **15** and accordingly any wire rupture is prevented. Mounting plate **18** furthermore comprises a feedthrough opening **18a** for the electric wiring **15**, which is electrically connected to the light emitting source **15**. Alternatively, power circuitry or power electronics (not depicted in the Figures) can be mounted (in the form of a PCB) to the mounting plate **18** and electrically connected with the electric wiring **15**. In this alternative example, the light emitting source **15** is subsequently connected with and powered by the power electronics PCB.

(15) The power circuitry or power electronics can also be mounted as a PCB inside the supporting structure **11**.

(16) Reference numeral **21** denotes a cup-shaped tensioning element, which is accommodated between the ball-joint structure **12** and the mounting plate **18**. The cup-shaped tensioning element **21** has an outer dimension which is smaller than the inner dimension of the housing **13** allowing it to be accommodated inside the lamp housing **13**. Due to the retaining spring element **19** being fitted in the circumferential slot **13d** provided in the inner circumferential surface **13c** of the housing **13**, the mounting plate **18** together with the light emitting source **14** as well as the cup-shaped tensioning element **21** are accommodated in a clamped manner between the ball-joint

surface **12** and the retaining spring element **19** and cannot dislodge from the inner space **130** of the housing **13**.

(17) The cup-shaped tensioning element **21** has an inner space **210** and a circumferential rim **21c**, which abuts against the ball surface of the ball-joint structure **12**. A feedthrough opening **21a** is present in the cup-shaped tensioning element **21** that links with the inner space **210** to allow passage of the electric wiring **15** via the several feedthrough openings **11a-12a** of the supporting structure **11** and the ball-joint structure **12** and via the feedthrough opening **18a** towards the light emitting source **15**.

(18) The clamping of the cup-shaped tensioning element **21** with its rim **21c** on the ball surface of the ball-joint structure **12** is effected by the retaining spring element **19**. Additionally, a tensioning spring **20** is mounted between the cup-shaped tensioning element **21** and the mounting plate **18**. Hereto, the cup-shaped tensioning element **21** is provided with a circumferential slit **21b** for accommodating the tensioning spring **20**. The slit **21b** is provided at the side of the cup-shaped tensioning element **21** facing towards the light exit window **13b** (or facing away from the ball-joint structure **12**).

(19) Similarly, the mounting plate **18** is provided with a circumferential slit **18c** for accommodating the other side of the tensioning spring **20**. The circumferential slit **18c** of the mounting plate **18** is provided in the side of the mounting plate **18** facing towards the cup-shaped tensioning element **21** (or facing away from the light exit window **13b**). Both circumferential slits **21b** and **18c** form a circumferential space enclosing the tensioning spring **20**. Herewith a compact inner construction is obtained with limited risk for malfunction or blocking.

(20) The mounting plate **18** and the cup-shaped tensioning element **21** are retained in the inner space **130** of the lamp housing **13** between the retaining spring element **19** and the ball-joint surface **12**. This construction together with the tensioning spring **20** mounted between the mounting plate **18** and the cup-shaped tensioning element **21** guarantees that a certain constant tension is maintained on the ball joint structure **12**. Accordingly, this construction allows the lamp housing **13** to be kept in the chosen position relative to the supporting structure **11**, whilst allowing rotation and orientation of the lamp housing **13** along the ball surface of the ball-joint structure **12** in any desired position.

(21) The assembly of the mounting plate **18** with the light emitting source **14**, the tensioning spring **20** and the cup-shaped tensioning element **21** is loosely accommodated within and relative to the lamp housing **13** between the retaining spring element **19** and the ball-joint structure **12**. The whole assembly of the parts **18(14)-20-21** can be kept stationary relative to the lamp housing **13**, whilst the latter rotation and directing in all directions around the ball surface of the ball joint structure **12** does not cause any twisting of the electric wire **15**. Accordingly any wire rupture is prevented.

(22) The electric wiring **15** is properly accommodated within the lamp housing **13** with the use of a flexible wiring sleeve **16** made from a flexible, yet sturdy material. The electric wiring **15** is shielded inside the flexible wiring sleeve **16**. The flexible wiring sleeve **16** has a first sleeve end **16a** which is loosely coupled with the ball-joint structure **12** and a second sleeve end **16b** which is loosely coupled with the mounting plate **18**. In particular, each first and second sleeve end **16a-16b** is provided with a coupling end **17a** and **17b** respectively, which is made from a sturdy, preferably plastic material.

(23) Each coupling end **17a-17b** has an outer dimension which is conformal to the inner dimension of the feedthrough opening **12a** (**18a**) of the ball-joint structure **12** (mounting plate **18**), respectively. In particular, each coupling end **17a-17b** has a coupling surface that loosely interacts with either the inner surface of the mounting flange **12b** surrounding the feedthrough opening **12a** of the ball-joint structure **12** or the inner surface of the mounting flange **18b** surrounding the feedthrough opening **18a** of the mounting plate **18**.

(24) The loose coupling of the flexible wiring sleeve **16** (with the coupling ends **17a-17b**) within the directional lighting device **10** prevents any twisting or entrapment during the rotation and

orientation of the lamp housing **13** along the ball-joint structure **12** relative to the supporting structure **11** and the light emitting device **14**.

LISTING OF REFERENCE NUMERALS USED

(25) **10** directional lighting device **11** supporting structure **11a** feedthrough opening of supporting structure **11b** mounting flange of supporting structure **12** ball-joint structure **12a** feedthrough opening of ball-joint structure **12b** mounting flange of ball-joint structure **12z** main axis or source axis of ball-joint structure **120** inner space of ball-joint structure **13** housing **13a** mounting end of housing **13b** light exit end of housing **13c** inner wall of housing **13d** circumferential slot in inner wall of housing **13z** longitudinal axis or destination axis of housing **130** inner space of housing **14** light emitting source **15** electric wiring **16** flexible wiring sleeve **16a-16b** first and second sleeve ends of flexible wiring sleeve **17a-17b** coupling ends for flexible wiring sleeve **18** mounting plate for light emitting source **18a** feedthrough opening of mounting plate **18b** mounting flange of mounting plate **18c** circumferential slit of mounting plate **19** retaining spring element **20** tensioning spring element **21** cup-shaped tensioning element **21a** feedthrough opening of cup-shaped tensioning element **21b** circumferential slit of cup-shaped tensioning element **21c** rim of cup-shaped tensioning element **210** inner space of cup-shaped tensioning element **22** bearing ring

Claims

1. A directional lighting device comprising: a supporting structure; a ball-joint structure mounted to the supporting structure; a housing configured to accommodate a light emitting source; electric wiring fed from the supporting structure through the ball-joint structure and electrically connecting the light emitting source; the ball-joint structure configured to direct the housing relative to the supporting structure and the light emitting source, wherein the light emitting source is mounted to a mounting plate, wherein the mounting plate has an outer dimension which is smaller than the inner dimension of the housing, and wherein a cup-shaped tensioning element having an outer dimension which is smaller than the inner dimension of the housing is clamped between the ball-joint structure and the mounting plate.
 2. The directional lighting device according to claim 1, wherein the housing is configured as an elongated hollow tube with one tube end being mounted to the ball-joint structure.
 3. The directional lighting device according to claim 2, wherein a bearing ring is mounted between the ball joint structure and the inner surface of the one tube end.
 4. The directional lighting device according to claim 1, wherein the mounting plate is retained within the housing by means of a retaining spring element, which fits in a circumferential slot provided at the inner circumferential surface of the housing.
 5. The directional lighting device according to claim 1, wherein a tensioning spring is mounted between the cup-shaped tensioning element and the mounting plate.
 6. The directional lighting device according to claim 5, wherein the cup-shaped tensioning element is provided with a circumferential slit for accommodating the tensioning spring.
 7. The directional lighting device according to claim 1, further comprising a flexible wiring sleeve for accommodating the electric wiring, wherein the flexible wiring sleeve has a first sleeve end loosely coupled with the ball-joint structure and a second sleeve end loosely coupled with the mounting plate.
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